Malik Choset

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Education

Carnegie Mellon University — Pittsburgh, PA
B.S. in Mechanical Engineering | GPA: 4.00 — *May 2029*
Relevant Coursework: Dynamics (24-351), Thermodynamics (24-221), Mechanics I: 2D Design (24-261), Electronics for Sensing & Actuation (24-251), Fundamentals of Programming (15-112), Manual Machining (24-203)

Experience

Allen Control Systems — Austin, TX
Mechanical Engineering Intern — *May 2026 – August 2026*

- Designed an Acceptance Test Procedure (ATP) mechanism to verify counter-drone systems prior to deployment.
- Optimized test rig for mass production using DFMA principles, reducing machining & assembly line bottlenecks.
- Created ASME Y14.5-compliant engineering drawings to ensure rapid, accurate component fabrication.
- Performed tolerance stack-up analyses on critical interfaces to ensure parts met exact dimensional specifications.

Carnegie Mellon Racing — Pittsburgh, PA/Remote
Rear Wing System Lead — *May 2026 – Present*

- Designed an adjustable Rear Wing system in SolidWorks for variable element configurations.
- Developed parameterized CFD models to automate the analysis pipeline and reduce setup time.
- Evaluated hundreds of configurations in CFD using geometric sweeps to find optimal designs.
- Optimized structural integrity using Ansys Mechanical, minimizing mass while meeting structural goals.
- Fabricated lightweight aerodynamic elements utilizing carbon fiber composite manufacturing techniques.
- Programmed CAM toolpaths and operated CNC machines to manufacture foam molds for carbon fiber layups.

Drag Reduction System Lead — *Sept. 2025 – May 2026*

- Designed the active aerodynamics element for Formula Student Electric race car utilizing SolidWorks and Ansys Fluent, achieving up to a 45% reduction in drag between open and closed positions.
- Validated system actuation by creating FBDs and using MATLAB to confirm performance goals.
- Optimized system structural performance using Ansys Mechanical, reducing mass by 75% over previous year.

CMU Department of Mechanical Engineering — Pittsburgh, PA
Lab TA, Fundamentals of Mechanical Engineering (24-101) — *August 2026 – Present*

- Led lab office hours and developed educational materials for undergraduate mechanical engineering students.
- Served as lead for a controls system project within the course curriculum.

Robotics Institute — Pittsburgh, PA
Modular Robotics Research Assistant — *June 2025 – August 2025*

- Developed a compact 2DoF module for the EigenBot system by optimizing motor packaging and material selections using Fusion 360, reducing the overall module form factor by 20%.
- Developed a Hall-Effect force-sensing gripper and foot module for the EigenBot modular robot system, driven by FEA results from Ansys Mechanical.

Fox Chapel Area School District — Pittsburgh, PA
Summer Custodian — *June 2022 – August 2024*

Projects

Cycloidal Drive Generator · *Python, SolidWorks API* — April 2026

- Developed an interactive app for designing cycloidal drives, allowing users to define certain parameters.
- Prevented invalid gear geometry by implementing parameter-checking algorithms.
- Automated the 3D modeling of gear components using the SolidWorks API, eliminating CAD modeling time.
- Automated a manufacturing pipeline by scripting the export from SolidWorks, creating 3D printable files.

Technical Skills

Mechanical Design & Analysis: DFM, DFA, FEA, GD&T, Tolerance Stack-up Analysis
Design Software: SolidWorks, Ansys SpaceClaim, Ansys Discovery, Fusion 360
Analysis Software: Ansys Mechanical, Ansys Fluent, Luminary CFD
Manufacturing: CNC Mill, Manual Mill, Manual Lathe, 3D Printing
Data Analysis: Python, Matlab, Microsoft Excel